

Pseudocode:

1. start
2. SET capA = 20, capB = 40, capC = 15
3. SET occA = 0, occB = 0, occC = 0
4. SET parkedCars = 0, parkedBikes = 0, parkedVans = 0
5. SET totalAccepted = 0, totalRejected = 0
6. REPEAT
7. Ask the user: "Enter number of expected vehicles to process"
8. Read input and store it in expectedVehicles
9. UNTIL expectedVehicles > 0
10. FOR i = 1 TO expectedVehicles DO
11. REPEAT
12. Ask the user: "Vehicle Type (C = Car, B = Bike, V = Van)"
13. Read input and store it in type
14. Convert type to uppercase
15. UNTIL type IS 'C' OR 'B' OR 'V'
16. REPEAT
17. Ask the user: "User Category (F = Faculty, S = Student, G = Guest/Visitor)"
18. Read input and store it in category
19. Convert category to uppercase
20. UNTIL category IS 'F' OR 'S' OR 'G'
21. REPEAT
22. Ask the user: "Valid Parking Permit? (Y/N)"
23. Read input and store it in permit
24. Convert permit to uppercase
25. UNTIL permit IS 'Y' OR 'N'
26. SET emergency = 'N'
27. IF permit IS 'N' THEN
28. REPEAT
29. Ask the user: "Is this an Emergency Vehicle? (Y/N)"
30. Read input and store it in emergency
31. Convert emergency to uppercase
32. UNTIL emergency IS 'Y' OR 'N'
33. ENDIF
34. IF permit IS 'N' AND emergency IS 'N' THEN
35. Display: "REJECTED: Vehicle has no valid permit and is not an emergency vehicle."
36. SET totalRejected = totalRejected + 1
37. CONTINUE to next vehicle
38. ENDIF
39. SET isParked = 0
40. IF category IS 'F' THEN
41. IF occA + 1 <= capA THEN
42. SET occA = occA + 1
43. SET isParked = 1
44. Display: "SUCCESS: Assigned to Zone A..."
45. ELSE
46. Display: "REJECTED: Zone A has no available space."

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47.     ENDIF
48.     ELSE IF category IS 'S' THEN
49.         IF type IS 'V' THEN
50.             IF occC + 2 <= capC THEN
51. SET occC = occC + 2
52.         SET isParked = 1
53.         Display: "SUCCESS: Student Van redirected to Zone C..."
54.         ELSE
55.         Display: "REJECTED: Zone C lacks 2 spaces..."
56.         ENDIF
57.         ELSE
58.         IF occB + 1 <= capB THEN
59.         SET occB = occB + 1
60.         SET isParked = 1
61.         Display: "SUCCESS: Assigned to Zone B..."
62.         ELSE
63.         Display: "REJECTED: Zone B has no available space."
64.         ENDIF
65.         ENDIF
66.         ELSE IF category IS 'G' THEN
67.         IF type IS 'V' THEN SET spacesNeeded = 2 ELSE SET spacesNeeded
           = 1 ENDIF
68.         IF occC + spacesNeeded <= capC THEN
69.         SET occC = occC + spacesNeeded
70.         SET isParked = 1
71. Display: "SUCCESS: Assigned to Zone C..."
72.         ELSE
73.         Display: "REJECTED: Zone C has no space."
74.         ENDIF
75.         ENDIF
76.         IF isParked IS 1 THEN
77.         SET totalAccepted = totalAccepted + 1
78.         IF type IS 'C' THEN SET parkedCars = parkedCars + 1

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79.         ELSE IF type IS 'B' THEN SET parkedBikes = parkedBikes + 1
80.         ELSE IF type IS 'V' THEN SET parkedVans = parkedVans + 1
81.         ENDIF
82.         ELSE
83.         SET totalRejected = totalRejected + 1
84.         ENDIF
85.         ENDFOR
86.         Display summary report and zone statistics
87.         End

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Algorithm:

1. Initialize capacities: capA = 20, capB = 40, capC = 15 and occupancies/counters to 0.
2. Ask the user to enter expectedVehicles until a positive integer is provided.
3. For each vehicle from 1 to expectedVehicles:
 - a. Ask the user to enter type ('C', 'B', 'V') until valid.
 - b. Ask the user to enter category ('F', 'S', 'G') until valid.
 - c. Ask the user to enter permit ('Y', 'N') until valid.
 - d. If permit is 'N', ask the user to enter emergency ('Y', 'N') until valid.
 - e. If permit is 'N' and emergency is 'N', reject the vehicle, increment totalRejected, and move to the next vehicle.
 - f. If category is 'F':
 - If $\text{occA} + 1 \leq \text{capA}$, assign to Zone A, update occA, set isParked = true.
 - Else, reject vehicle.
 - g. Else if category is 'S':
 - If type is 'V':
 - If $\text{occC} + 2 \leq \text{capC}$, assign to Zone C, update occC, set isParked = true.
 - Else, reject vehicle.

- Else:
 - If $\text{occB} + 1 \leq \text{capB}$, assign to Zone B, update occB, set isParked = true.
 - Else, reject vehicle.
 - h. Else if category is 'G':
 - Determine spaces needed (2 if 'V', else 1).
 - If $\text{occC} + \text{spacesNeeded} \leq \text{capC}$, assign to Zone C, update occC, set isParked = true.
 - Else, reject vehicle.
 - i. If isParked is true:
 - Increment totalAccepted and increment specific vehicle count (parkedCars, parkedBikes, or parkedVans).
 - j. Else, increment totalRejected.
4. Determine the highest occupied zone among Zone A, Zone B, and Zone C.
 5. Display parking summary report including totals, breakdown by vehicle type, zone occupancies, highest occupied zone, and full status.

Question 5:

IPO Chart

<u>INPUT</u>	<u>PROCESSING</u>	<u>MODULE REFERENCE</u>	<u>OUTPUT</u>
expectedVehicles	1. Enter expected number of vehicles	Read	Individual vehicle assignment status
type	2. Loop for each vehicle: a. Enter vehicle type, category, permit, emergency b. Validate inputs and convert to uppercase c. Check eligibility (Permit / Emergency) d. Assign zone based on user category & vehicle type e. Update zone occupancy and counts	Read Validate CheckEligibility AssignZone UpdateCounts	Total vehicles processed
category	3. Calculate summary statistics and highest occupancy zone	Calc	Total accepted vehicles
permit	4. Print parking summary report	Print	Total rejected vehicles
emergency	5. End	ParkingControl	Parked cars count
			Parked bikes count
			Parked vans count

			Zone occupancy & remaining capacity
			Highest occupancy zone
			Campus parking status

PAC Table:

Given Data	Required Result	Processing Required	Solution Alternative
Constant capacities: capA = 20, capB = 40, capC = 15	Individual vehicle decision message (Success/Rejected)	Read and validate expectedVehicles (> 0)	Use do-while loops for input validation
Total expected vehicles (expectedVehicles)	Assigned parking zone and updated remaining space	Loop through each vehicle from 1 to expectedVehicles	Use conditional if-else blocks for zone allocation and capacity checks
Vehicle type (type: C, B, V)	Total count of processed, accepted, and rejected vehicles	Read and validate type, category, and permit	Use a for loop to iterate through the specified number of vehicles

- User category (category: F, S, G)	Total counts of parked cars, bikes, and vans	Read emergency if permit == 'N'	Alternative: Divide validation, allocation, and reporting into separate modular functions
Parking permit status (permit: Y, N)	Zone occupancy breakdown (occA, occB, occC) and remaining capacities	Reject vehicle if permit == 'N' and emergency == 'N'	
Emergency status (emergency: Y, N)	Identification of the highest occupancy zone	Allocate Zone A for Faculty ('F') if $\text{occA} + 1 \leq \text{capA}$	
	Campus parking status (FULL or Available Space)	Allocate Zone C (2 spaces) for Student ('S') Van if $\text{occC} + 2 \leq \text{capC}$	
		Allocate Zone B for Student ('S') non-Van if $\text{occB} + 1 \leq \text{capB}$	
		Allocate Zone C for Guest ('G') based on type (2 spaces for Van, 1 for others)	
		Increment total accepted and	

		individual vehicle type counts if parked	
		Increment total rejected count if not parked	
		Calculate maximum zone occupancy between occA, occB, and occC	
		Check if all zones are fully occupied (occA == capA, occB == capB, occC == capC)	

